



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

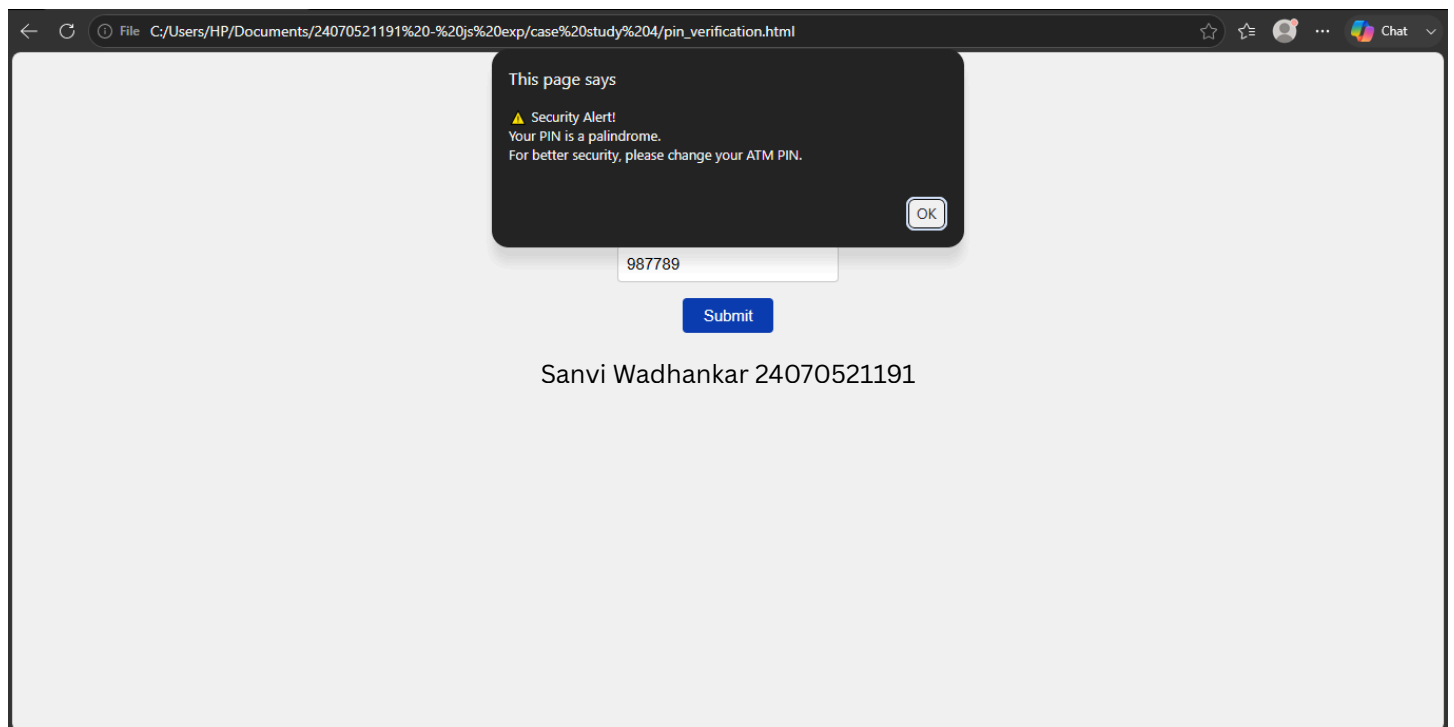
Case-Study 4

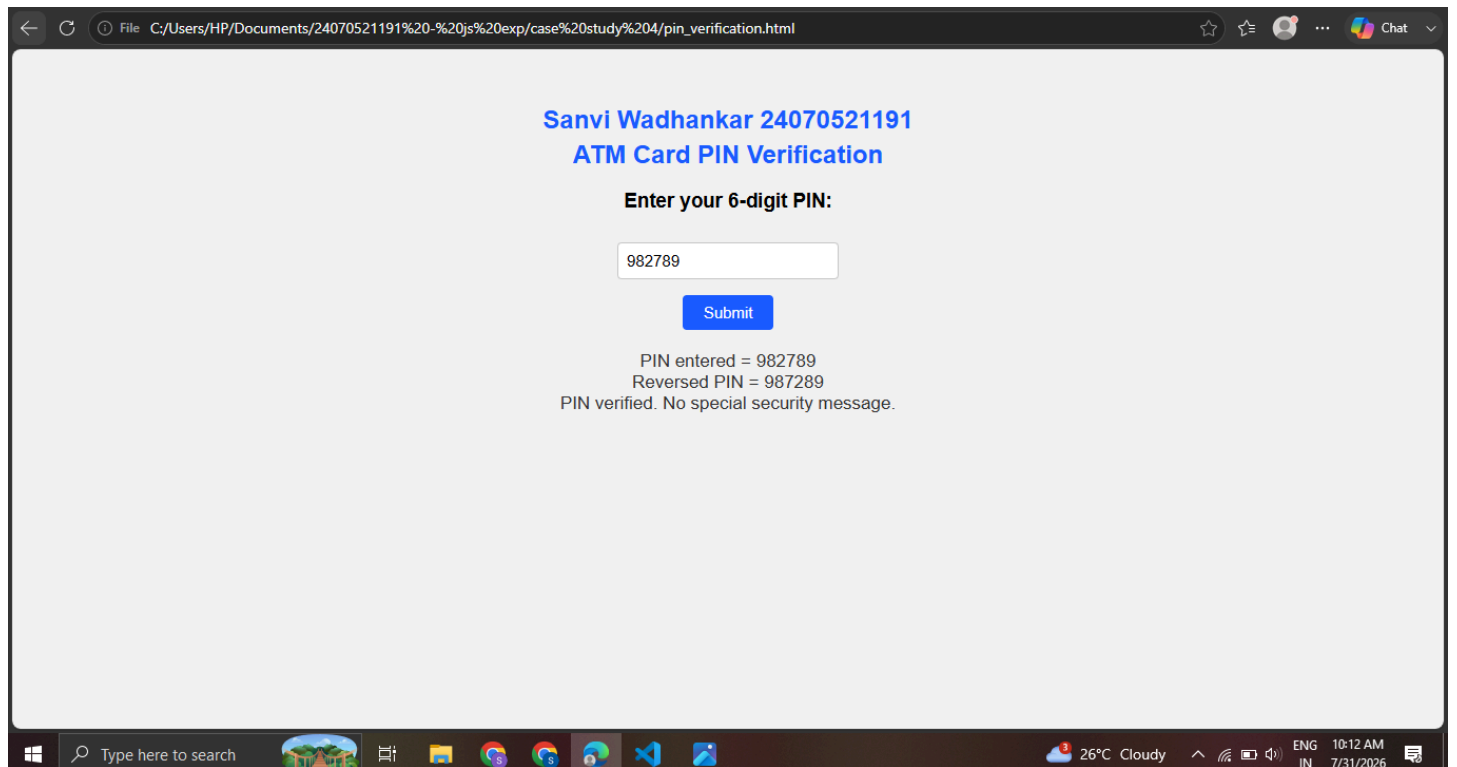
Real-Time Case Study: ATM Card PIN Verification Using JavaScript Functions

Case Study

An ATM system verifies whether a customer's PIN is a palindrome. If the PIN reads the same forward and backward, the bank displays a special security message. To implement this, create a function to reverse the entered PIN number and another function to check whether it is a palindrome. Use function declaration, function expression, arrow function, and demonstrate scope concepts.

Output:-





Theory:-

- A palindrome is a word, phrase, number, or sequence of characters that remains identical when read in reverse.
- The try...catch statement in JavaScript wraps a block of code to test for runtime errors and handle them without crashing your entire application.

```
try { // Code that might throw an error
```

```
} catch (error) { // Code that runs only if an error occurs
```

```
} finally { // Code that always runs (optional)
```

```
}
```

- **Scope** determines where variables can be accessed in your code, while a **closure** is a mechanism where an inner function permanently remembers and accesses variables from its outer (parent) scope, even after the outer function has finished executing

Code:-

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Palindrome Checker</title>
```

```
<style>

  body {

    font-family: Arial, sans-serif;

    background: #f4f4f4;

    text-align: center;

    margin-top: 50px;

  }

  .container {

    width: 350px;

    margin: auto;

    background: white;

    padding: 20px;

    border-radius: 10px;

    box-shadow: 0 0 10px gray;

  }

  input {

    width: 90%;

    padding: 10px;

    margin: 10px;

  }

  button {

    padding: 10px 20px;

    cursor: pointer;

  }

  h3 {

    color: blue;

  }

</style>

</head>
```

```
<body>
```

```
<div class="container">
```

```
<h2>Sanvi Wadhankar 24070521191</h2>
```

```
<h2>Palindrome Checker</h2>
```

```
<input type="text" id="word" placeholder="Enter a word">
```

```
<button onclick="checkPalindrome()">Check</button>
```

```
<h3 id="result"></h3>
```

```
</div>
```

```
<script>
```

```
// Global Scope
```

```
let appName = "Palindrome Checker";
```

```
// Function Declaration
```

```
function cleanString(str) {
```

```
    return str.toLowerCase().replace(/[^a-z0-9]/g, "");
```

```
}
```

```
// Function Expression
```

```
const reverseString = function(str) {
```

```
    return str.split("").reverse().join("");
```

```
};
```

```
// Closure
```

```
function counter() {
```

```
    let count = 0;
```

```
    return function() {
```

```
        count++;

        return count;

    };

}
```

```
const checkCount = counter();
```

```
// Arrow Function
```

```
const isPalindrome = (str) => {

    return str === reverseString(str);

};
```

```
// Main Function
```

```
function checkPalindrome() {

    try {

        let input = document.getElementById("word").value;

        // Local Scope

        let text = cleanString(input);

        if (text === "") {

            throw "Please enter some text.";

        }

        let result = isPalindrome(text)

        ? `${input}" is a Palindrome 
```

```
    } catch(error) {

        document.getElementById("result").innerHTML =

            `<span style="color:red;">Error: ${error}</span>`;

    }

    finally {

        console.log(appName + " executed.");

    }

}

</script>

</body>

</html>
```

Conclusion:-

Hence we have created a pin verification system for ATM that gives alert if the pin entered is a palindrome telling that the password is weak and needs to be changed.